Cross-Lingual Knowledge Conflict Signals Inherit Directional Asymmetries from General Multilingual Representations

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Abstract

When multilingual large language models (LLMs) encounter conflicting information across languages, presentation language can determine the answer. Behavioral studies attribute this to linguistic affinity, but the internal mechanism remains unknown: how is the conflict signal encoded, does it share structure with general cross-lingual representations, and does detectability imply causal influence on resolution?

We probe residual stream representations in Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-2-9B-IT, focusing on English–Chinese conflicts. Our analysis combines cross-lingual transfer, XNLI non-conflict baselines, confound exclusion, and representation steering. Conflict signals are linearly detectable across languages, and cross-lingual transfer exhibits a directional asymmetry: probes trained on Chinese transfer to English ∼7 percentage points more accurately than the reverse. This asymmetry is consistent with the XNLI non-conflict baseline in both magnitude and direction, with overlapping confidence intervals, inheriting from the model's general multilingual structure rather than conflict-specific encoding. A dissociation emerges between detection and decision in Llama-3.1-8B-Instruct: the probe signal peaks at late layers (88% depth), but representation steering achieves causal influence only at a mid-layer (41% depth); analogous Qwen and Gemma steering yields no significant effect. These findings demonstrate that linear detectability does not imply causal relevance, with implications for probe-based intervention research.

1 Introduction

When multilingual large language models (LLMs) encounter conflicting information across languages, the language of presentation can determine which answer the model produces. Zhao et al. (2026) showed that linguistic affinity dominates conflict resolution in entity-centric factual conflicts, with models preferring answers from linguistically closer sources. Yet the internal mechanism behind this behavior remains opaque: how does the model encode the conflict signal? Can it be detected from internal representations? And if detectable, does the detected signal causally influence the model's resolution strategy? These questions sit at the intersection of mechanistic interpretability and multilingual NLP, and answering them requires moving beyond behavioral observations to probe model internals directly.

Existing work on probing knowledge conflict signals has been limited to monolingual settings. Zhao et al. (2024) showed that residual stream representations carry linearly detectable conflict signals in English, and Brink et al. (2026) performed knowledge attribution probing to localize factual associations within model parameters. Extending these approaches cross-lingually introduces additional challenges. Translation quality may confound probe transfer: a poor translation could alter conflict detection for reasons unrelated to the conflict signal itself. Token overlap between languages, even when minimal, could provide a shortcut for cross-lingual probes. Observed asymmetries in transfer accuracy may reflect general cross-lingual properties rather than conflict-specific phenomena, and disentangling these sources requires careful baseline comparisons. Beyond detectability, a fundamental limitation of probing is that it establishes correlation, not causation (Hewitt and Liang, 2019). Basu et al. (2026) demonstrated that even near-perfect probe accuracy does not guarantee that the detected direction can steer model behavior.

We address these questions through a pipeline that combines residual stream probing with representation steering across three models of distinct architectures: Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-2-9B-IT. Probes achieve 77.6%,

75.0%, and 76.2% balanced accuracy on English within-language conflict detection, respectively, and all cross-lingual transfer directions exceed chance (binomial $p < 0.001$). Cross-lingual transfer exhibits a directional asymmetry: probes trained on Chinese transfer to English approximately 7 percentage points (pp) more accurately than the reverse in all three models. This asymmetry, however, is not conflict-specific. XNLI entailment-versus-contradiction probes (Conneau et al., 2018) show the same directional pattern at comparable magnitude (+6.0, +6.7, and +7.8 pp), with substantially overlapping confidence intervals. The conflict asymmetry thus inherits from a general property of English-dominant models, where Chinese representations project more readily onto English representational space than the reverse. Our causal analysis reveals a dissociation between detection and decision in Llama-3.1-8B-Instruct: the probe signal peaks at late layers (L28, 88% depth), but representation steering succeeds only at a mid-layer (L13, 41% depth; Fisher $p = 0.002$), with all late-layer interventions producing no significant behavioral change. To ensure these findings are robust, we apply a confound exclusion pipeline that includes COMET-based translation quality stratification, token overlap analysis, and non-conflict baseline comparison.

These results carry two methodological implications. The layer dissociation between probing peak and causal influence demonstrates that the layer with highest probe accuracy is not necessarily where the model resolves the conflict. This challenges the common practice of intervening at the probing peak layer, consistent with the warning from Basu et al. (2026) that detectability and causal relevance can diverge. The directional asymmetry in cross-lingual transfer reflects a structural property of English-dominant multilingual models, not a conflict-specific phenomenon. Any cross-lingual probing study must account for this baseline asymmetry to correctly interpret transfer differences.

Our contributions are as follows:

- **Cross-lingual conflict detectability and asymmetry characterization.** We provide the first evidence that knowledge conflict signals are linearly detectable across languages in residual stream representations. The transfer asymmetry (ZH→EN > EN→ZH by ∼7 pp) matches the XNLI non-conflict baseline in both magnitude and direction, establishing that it reflects the model's general cross-lingual structure rather than conflict-specific encoding.

- **Detection-decision layer dissociation.** In Llama-3.1-8B-Instruct, representation steering at the mid-layer (L13, 41% depth) causally shifts conflict resolution (Fisher $p = 0.002$), while steering at the late-layer probing peak (L28, 88% depth) produces no significant effect. This dissociation shows that linear detectability does not imply causal relevance, and that conflict resolution occurs upstream of where the outcome becomes linearly readable.

- **Confound exclusion pipeline for cross-lingual probing.** We design and validate a systematic pipeline of COMET translation quality ablation, token overlap analysis, and non-conflict task baseline comparison that excludes common confounds in cross-lingual probing studies.

2 Related Work

Knowledge conflict detection. Xie et al. (2024) introduced ConflictQA and showed that models exhibit context-dependent resolution strategies, and Xu et al. (2024) provided a systematic conflict taxonomy (intra-memory, context-memory, inter-context). Moving beyond behavioral observation, Zhao et al. (2024) demonstrated linearly separable conflict signals in English residual streams, Brink et al. (2026) localized factual associations via probing, Pham et al. (2026) traced intra-memory conflict pathways, and Li et al. (2025) proposed conflict mitigation methods. These studies operate exclusively in monolingual settings; extending to cross-lingual scenarios introduces confounds (translation quality, token overlap, general transfer properties) that we address with a dedicated exclusion pipeline.

Cross-lingual representations in multilingual models. Intermediate layers develop language-independent representations: Wu et al. (2020) showed cross-lingual structure emerges without explicit alignment, and Muller et al. (2021) found mid-layer alignment in multilingual BERT. For decoder-only LLMs, Wendler et al. (2024) showed that Llama processes non-English inputs through an English-centric latent space, and Kargaran et al. (2024) quantified this bias across language pairs.

These findings predict our empirical observations: mid-layer steering success aligns with intermediate-layer language alignment, and the ZH→EN transfer advantage reflects English-centric structure. Behavioral consequences have been documented by Wang et al. (2025), Zhao et al. (2026) (CLEAR benchmark), and Ifergan et al. (2024); Xing et al. (2024); our work provides a mechanistic account of why such asymmetries arise.

Representation steering and causal intervention. Li et al. (2023) introduced Inference-Time Intervention for steering via learned bias vectors, Turner et al. (2023) proposed Activation Addition as an optimization-free alternative, and Zou et al. (2023) generalized these approaches through Representation Engineering. Marks and Tegmark (2024) showed that truth-related representations exhibit linear structure amenable to such interventions. In knowledge conflicts, Zhao et al. (2025) applied SAE-based steering (SpARE) in monolingual settings. These successes are not universal: Basu et al. (2026) showed that near-perfect probe accuracy on clinical NLP tasks failed to translate into effective steering. Feng et al. (2026) proposed fine-grained activation steering targeting specific layers, and Meng et al. (2022) demonstrated localized causal editing via activation patching (ROME); alternative causal frameworks (Geiger et al., 2024) remain unexplored for cross-lingual conflict settings. Our results identify layer selection as a critical variable: mid-layer intervention succeeds where late-layer intervention fails, suggesting conflict resolution occurs upstream of where the outcome becomes linearly readable.

3 Method

Our pipeline proceeds from data construction through probing and steering, formalized below.

3.1 Task Formulation

Given a context passage c and question q, the model produces an answer that is either *contextual* (derived from c) or *parametric* (from pretrained knowledge). When c contradicts parametric knowledge, the choice constitutes a knowledge conflict. We train a logistic regression probe $f_\theta(\mathbf{h}_l) = \sigma(\mathbf{w}_l^\top \mathbf{h}_l + b_l)$ on the last-token residual stream representation $\mathbf{h}_l \in \mathbb{R}^d$ at layer l to predict a binary label $y \in \{\text{contextual}, \text{parametric}\}$.

3.2 Data Construction

We build on ConflictQA (Xie et al., 2024), which pairs questions with context passages contradicting the model's parametric knowledge. We filter samples by querying each model without context and retaining only those answered correctly, yielding 1,368 samples for Llama (Grattafiori et al., 2024), 966 for Qwen (Yang et al., 2025), and 1,590 for Gemma (Gemma Team et al., 2024) (80/20 train/test split with stratification). For cross-lingual transfer, we translate all inputs into Chinese using a Qwen3-8B pipeline, with quality validated by COMET (Rei et al., 2020) back-translation scores used in confound analysis (§3.5).

3.3 Activation Extraction and Probing

For each input, we construct English and Chinese versions and extract the last-token residual stream $\mathbf{h}_l \in \mathbb{R}^d$ at every layer l ($L = 32$ for Llama, 36 for Qwen, 42 for Gemma; $d = 4096$ for Llama/Qwen, 3584 for Gemma), following the last-token aggregation principle (Belinkov, 2022). We record the model's answer and assign binary labels (contextual vs. parametric).

At each layer, we train an L_2-regularized logistic regression probe (Belinkov, 2022; Hewitt and Liang, 2019) and evaluate with balanced accuracy. We report peak-layer accuracy (maximum across layers) as the primary metric. Confidence intervals are estimated via bootstrap resampling ($n = 2{,}000$) (Ulmer et al., 2022).

3.4 Cross-Lingual Transfer Protocol

We evaluate probes in four directions: EN→EN, ZH→ZH (within-language upper bounds), EN→ZH, and ZH→EN (cross-lingual transfer). We quantify directional asymmetry as $\Delta_{\text{asym}} = \text{acc}(\text{ZH}\to\text{EN}) - \text{acc}(\text{EN}\to\text{ZH})$, computed via paired bootstrap (same resampled indices for both directions).

3.5 Confound Exclusion

We exclude two artifacts: (1) *translation quality*—we stratify test samples by COMET score into terciles and compare probe accuracy (Fisher's exact test); (2) *token overlap*—we compute per-sample Jaccard similarity between English and Chinese token sets and test for correlation with accuracy (Pearson's r). As a *non-conflict baseline*, we apply the same pipeline to XNLI entailment-versus-contradiction (Conneau et al., 2018); if $\Delta_{\text{asym}} > 0$

also appears there, the asymmetry reflects a general cross-lingual property.

3.6 Representation Steering

To test whether the probe direction is causally relevant (Hewitt and Liang, 2019), we inject $\alpha \cdot \mathbf{w}_{\text{probe}}$ into the last-token residual stream at layer l during inference (Li et al., 2023; Zou et al., 2023): $\mathbf{h}'_l = \mathbf{h}_l + \alpha \cdot \mathbf{w}_{\text{probe}}$, where positive α pushes toward the contextual direction and negative toward the parametric. As control, we inject a random vector of equal norm. We measure the flip rate (fraction of samples switching answer category) and compare via Fisher's exact test. We apply Bonferroni correction per layer ($k = 10$ α values), treating each layer as an independent hypothesis about causal locus—consistent with prior work in mechanistic interpretability that tests layers individually (Meng et al., 2022; Geiger et al., 2024). For Llama L13 (the layer showing causal effects), this yields a corrected threshold of $0.05/10 = 0.005$. For last-token injection, we sweep $\alpha \in \{\pm 20, \pm 10, \pm 5, \pm 2, \pm 1\}$ ($k = 10$) at three Llama layers (L13, L18, L28) spanning 41–88% depth; all-position and cross-model experiments use a reduced 6-value grid ($\alpha \in \{\pm 20, \pm 10, \pm 5\}$). Steering evaluation uses only the conflict-condition subset of the English test data, yielding n_{test} smaller than the full probe test set (e.g., 136 vs. 272 for Llama).

4 Experiments

4.1 Experimental Setup

Models. We probe three instruction-tuned models with comparable parameter counts but distinct architectures: Llama-3.1-8B-Instruct (32 layers, hidden dimension 4096) (Grattafiori et al., 2024), Qwen3-8B (36 layers, hidden dimension 4096) (Yang et al., 2025), and Gemma-2-9B-IT (42 layers, hidden dimension 3584) (Gemma Team et al., 2024).

Dataset. We use ConflictQA (Xie et al., 2024), a benchmark of context-parametric knowledge conflicts where a provided context contradicts the model's parametric knowledge. After verifying that each model can correctly answer the parametric-knowledge question without context, we retain 1,368 samples for Llama (1,096 train / 272 test), 966 for Qwen (773 train / 193 test), and 1,590 for Gemma (1,274 train / 316 test), split 80/20 with stratification.

Translation. We translate all English contexts and questions into Chinese using a Qwen3-8B translation pipeline (EN→ZH), with quality validated by COMET (Rei et al., 2020) back-translation scores.

Probing. For each layer l, we train a logistic regression probe on the last-token residual stream representation $\mathbf{h}_l$, with binary labels indicating whether the model chose the contextual or parametric answer. We report balanced accuracy to account for class imbalance. All confidence intervals (CIs) are computed via bootstrap resampling with paired differences for asymmetry estimation.

Non-conflict baseline. To test whether cross-lingual transfer patterns are specific to knowledge conflicts, we run the same probing pipeline on XNLI (Conneau et al., 2018), classifying entailment versus contradiction.

4.2 Within-Language Conflict Detection

Table 1 reports balanced accuracy for all probe transfer directions. All three models encode linearly detectable conflict signals in their residual streams. Llama achieves 77.6% on EN→EN (peak at layer 28, 88% depth) and 72.1% on ZH→ZH. Qwen reaches 75.0% on EN→EN (peak at layer 24, 67% depth) and 72.6% on ZH→ZH. Gemma attains 76.2% on EN→EN (peak at layer 24, 57% depth) and 71.0% on ZH→ZH. All within-language accuracies are significantly above chance (binomial $p < 0.001$). Figure 1 shows the layer-wise probe accuracy curves for all three models. These results confirm that knowledge conflict signals are linearly separable in the residual stream across all three architectures and both languages.

4.3 Cross-Lingual Transfer and Asymmetry

All four cross-lingual transfer directions achieve balanced accuracy above chance (binomial $p < 0.001$; Table 1), confirming that conflict representations partially overlap across languages. All three models show a consistent directional trend: probes trained on Chinese representations transfer to English (ZH→EN) more accurately than the reverse direction (EN→ZH). For Llama, $\Delta_{\text{asym}} = +7.3$ percentage points (pp) with a 95% bootstrap CI of [0.4, 14.0] ($p = 0.038$); for Qwen, $\Delta_{\text{asym}} = +7.2$ pp with CI [−0.6, 13.7]; for Gemma, $\Delta_{\text{asym}} = +6.6$ pp with CI [1.9, 11.1] ($p = 0.006$). While the Qwen CI marginally includes zero, the asymmetry

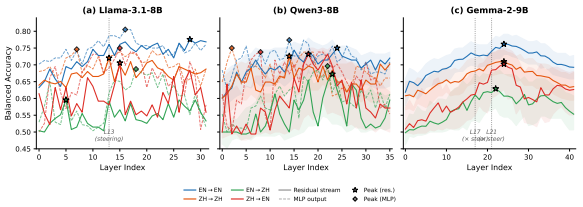


Figure 1: Layer-wise probe balanced accuracy for Llama-3.1-8B (left), Qwen3-8B (center), and Gemma-2-9B-IT (right). Solid lines: residual stream probes; dashed lines (Llama and Qwen only): MLP sublayer probes. Stars mark peak layers. In the Llama panel, the vertical annotation at L13 marks the layer where steering produces significant causal influence (Fisher $p = 0.002$), coinciding with the ZH→ZH probe peak but not the EN→EN peak (L28). In the Gemma panel, L17 and L21 are annotated as steering-negative layers.

Table 1: Single-seed balanced accuracy (%) at each direction's optimal layer. Δ_{asym} and 95% CIs from paired bootstrap ($n = 2{,}000$ resamples). Bold: best per-task direction.

Direction	Llama-3.1-8B	Qwen3-8B	Gemma-2-9B
Conflict Task			
EN → EN	**77.6**	**75.0**	**76.2**
ZH → ZH	72.1	72.6	71.0
EN → ZH	59.6	67.3	62.9
ZH → EN	70.6	73.2	70.2
Δ_{asym}	+7.3 [0.4, 14.0]	+7.2 [−0.6, 13.7]	+6.6 [1.9, 11.1]
XNLI Baseline			
EN → EN	94.0	96.6	95.6
ZH → ZH	87.5	89.5	89.0
EN → ZH	89.0	90.5	89.4
ZH → EN	**94.9**	**97.2**	**97.5**
Δ_{asym}	+6.0 [0.9, 11.3]	+6.7 [2.4, 11.7]	+7.8 [5.7, 10.0]

Note: Δ_{asym} is the mean of paired bootstrap differences, not the arithmetic difference of the reported single-seed accuracies.

is consistent in direction across all three architectures. The magnitude of this asymmetry is consistent in direction and magnitude with the XNLI non-conflict baseline across all three architectures (§4.4), and the consistent direction across all three models suggests a shared structural property rather than a model-specific artifact (Figure 2).

4.4 Non-Conflict Baseline Comparison

If the ZH→EN transfer advantage were specific to knowledge conflicts, it should be absent or substantially reduced in a non-conflict semantic task. We test this prediction using XNLI entailment-versus-contradiction probes. All three models achieve high within-language accuracy on XNLI (Llama 94.0% EN, 87.5% ZH; Qwen 96.6% EN, 89.5% ZH; Gemma 95.6% EN, 89.0% ZH), confirming that the probing pipeline captures semantic distinctions beyond knowledge conflicts. The XNLI probes exhibit the same directional asymmetry: Llama $\Delta_{\text{asym}} = +6.0$ pp, CI [0.9, 11.3]; Qwen $\Delta_{\text{asym}} = +6.7$ pp, CI [2.4, 11.7]; Gemma $\Delta_{\text{asym}} = +7.8$ pp, CI [5.7, 10.0] ($p < 0.001$). All three XNLI CIs exclude zero and overlap substantially with the corresponding conflict-task CIs (Figure 2). The consistency in asymmetry magnitude between conflict probes (Llama +7.3, Qwen +7.2, Gemma +6.6 pp) and XNLI probes (Llama +6.0, Qwen +6.7, Gemma +7.8 pp) indicates that the ZH→EN transfer advantage is not specific to knowledge conflict representations. Formal TOST equivalence tests (±2 pp margin) were non-significant for all three models (Llama $p = 0.434$, Qwen $p = 0.366$, Gemma $p = 0.377$), reflecting insufficient power (SE ≈ 4.37 pp) rather than meaningful differences—point estimates fall within ±1.3 pp. All six model×task combinations exhibit the same directional pattern (ZH→EN > EN→ZH; binomial $p = 0.016$), with magnitudes clustering in [+6.0, +7.8 pp]. The asymmetry thus reflects a general cross-lingual property, not a conflict-specific phenomenon.

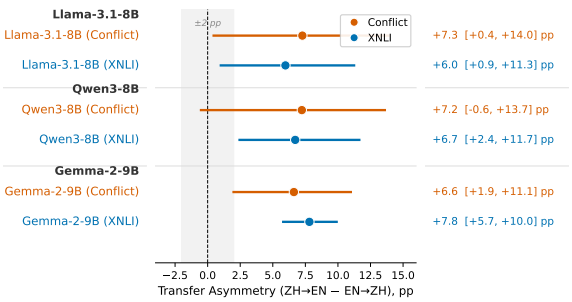


Figure 2: Transfer asymmetry (Δ_{asym}) with bootstrap 95% CIs ($n = 2{,}000$) for conflict and XNLI tasks across all three architectures (six data points). Conflict and XNLI CIs overlap substantially within each model, indicating that the ZH→EN advantage is a general cross-lingual property rather than conflict-specific. Grey band: ±2 pp equivalence zone.

4.5 Confound Exclusion

Two potential confounds could inflate cross-lingual probe accuracy: translation quality and lexical overlap between English and Chinese inputs.

Translation quality. We stratify test samples by COMET score into terciles (high, medium, low) and compare EN→ZH probe accuracy across strata. If translation quality drove probe transfer, high-COMET samples would yield higher accuracy. Ta-

Table 2: Confound exclusion analysis. COMET tercile: Δ = balanced accuracy difference between high and low COMET score terciles for EN→ZH probes (Fisher's exact test). Token overlap: Pearson correlation between per-sample Jaccard similarity and EN→ZH probe accuracy.

Confound	Model	Result	p
COMET tercile	Qwen	$\Delta = -5.6$ pp	0.695
COMET tercile	Llama	$\Delta = +1.7$ pp	0.761
Token overlap	Qwen	n.s.	>0.4
Token overlap	Llama	$r = -0.15$	0.012

ble 2 shows no such pattern: Qwen exhibits a reversed direction ($\Delta = -5.6$ pp, Fisher $p = 0.695$), and Llama shows a negligible effect ($\Delta = +1.7$ pp, Fisher $p = 0.761$). Translation quality does not confound the cross-lingual transfer results.

Token overlap. Per-sample Jaccard similarity between English and Chinese tokenized inputs averages 3–4%, consistent with the minimal lexical overlap between the two languages. Neither model shows a positive correlation between token overlap and probe accuracy: Qwen yields no significant correlation ($p > 0.4$), and Llama shows a weak negative correlation ($r = -0.15$, $p = 0.012$), where higher overlap is associated with *lower* accuracy. Shared tokens do not drive cross-lingual transfer. As additional evidence, we extend the token overlap analysis to EN-DE, where per-sample Jaccard similarity (14.8%) is approximately $4\times$ higher than EN-ZH (3.8%). Despite this substantially higher lexical overlap, the EN↔DE transfer asymmetry is near-zero ($\Delta = +1.2$ pp, n.s.), further ruling out shared tokens as a driver of the observed EN-ZH asymmetry.

4.6 Causal Evidence via Representation Steering

The probing results establish that conflict signals are detectable, but detection does not imply causal relevance. We test whether the probe weight direction $\mathbf{w}_{\text{probe}}$ causally influences conflict resolution by injecting $\alpha \cdot \mathbf{w}_{\text{probe}}$ into the last-token residual stream at a target layer during inference. We measure the flip rate (fraction of samples that switch answer category) and compare against a random-direction control of equal magnitude via Fisher's exact test.

L13 (mid-layer, 41% depth). Steering at layer 13 produces bidirectional causal effects (Table 3). Positive steering ($\alpha = +5$) yields a flip rate of 19.9% versus 6.6% for the random control (Fisher $p = 0.002$; 4 p→c flips), while negative steering ($\alpha = -20$) yields 36.0% versus 17.6% ($p < 0.001$; 5 c→p flips); both survive Bonferroni correction. All-position injection at L13 produces identical results, confirming that the causal signal is concentrated at the final token. Steering at nearby L16 (50% depth, just three layers away) showed no significant effect (all $p > 0.12$), indicating a remarkably narrow causal window. Under a conservative family-wise correction across all 30 tests (3 layers $\times$ 10 α values, threshold $p < 0.00167$), L13 remains significant ($\alpha = +10$, $p = 0.0003$). An independent, pre-planned boundary-mapping replication with separately sampled data ($N = 201$, six layers from L12 to L28) confirmed L13 significance ($p < 0.001$) and localized the causal boundary: L12 showed no significant effect ($p = 0.085$), with attenuated but significant effects at L14 ($p = 0.002$) and L16 ($p = 0.005$, suggesting the original L16 null may reflect limited power), diminishing to non-significance after Bonferroni correction at L18 ($p = 0.010$) and L28 ($p = 0.284$; Appendix Table 4), revealing a spatially precise causal gradient centered at L13.

Negative layers and coherence collapse. Steering at L28 (probe peak, 88% depth) produces no significant effect with either last-token or all-position injection (all $p > 0.3$); L18 (MLP peak, 56%) is similarly negative ($p > 0.44$). At extreme magnitude ($\alpha = +20$), the control flip rate exceeds steering ($p = 0.053$), indicating coherence collapse.

Detection ≠ Decision (Llama). In Llama-3.1-8B-Instruct, the detectable conflict signal peaks at L28 (88% depth), but causal influence is localized to L13 (41% depth): late layers encode the outcome in a linearly readable form, while mid-layers host the computation that determines it.

Cross-model steering. Qwen steering at L14 (39%, $n_{\text{test}} = 131$; probe accuracy 71.0%) produced no significant effect at any α (best $p = 0.785$) and zero directional flips. Gemma-2-9B-IT L17 (40% depth) provides a stronger test: with 88.9% probe accuracy—the highest among all three models—steering produced no significant change at any α (best $p = 0.448$) and zero directional flips

Table 3: Representation steering results. Flip rate: fraction of test samples switching answer category. Control: random-direction perturbation of equal magnitude. Bonferroni-corrected significance threshold $p < 0.005$ ($k = 10$). Significant results in bold.[†]

Layer	Depth	α	Steer (%)	Ctrl (%)	Fisher p	Dir.
L13	**41%**	**+5**	**19.9**	**6.6**	**0.002**	**4 p→c**
L13	**41%**	**−20**	**36.0**	**17.6**	**< 0.001**	**5 c→p**
L16	50%	+5	8.8	8.1	1.000	
L16	50%	+10	18.4	11.0	0.123	
L16	50%	+20	22.8	24.3	0.886	
L18	56%	+2	3.7	1.5	0.447	
L28	88%	−20	14.7	16.9	0.740	
L28 (all)	88%	−20	11.8	11.8	1.000	
Qwen3-8B ($n_{test} = 131$)						
L14	39%	best	n.s.		0.785	0 dir.
Gemma-2-9B-IT ($n_{test} = 171$)[‡]						
L17	40%	best	n.s.		0.448	0 dir.

[†] At $\alpha = +20$, the control flip rate exceeds steering ($p = 0.053$), indicating coherence collapse at extreme perturbation magnitudes. [‡] Gemma v2: re-run after correcting a label assignment bug in v1.

($n_{\text{test}} = 171$). Gemma's 88.9% probe accuracy—the highest among all three models, exceeding both Llama (76.1%) and Qwen (71.0%)—rules out probe quality as a confound for the negative steering result, confirming that the causal influence of probe-weight directions is architecture-dependent. Appendix C reports matched-condition results at three layers per model (∼36–43% depth), all confirming the null. Cross-task language probe transfer further validates this shared subspace (§5).

5 Discussion

Detection and decision occur at different layers. Our central mechanistic finding is a dissociation between where conflict signals become linearly readable and where they exert causal influence on model behavior. In Llama, probe accuracy on EN→EN peaks at layer 28 (88% depth), yet representation steering succeeds only at layer 13 (41% depth; Fisher $p = 0.002$), and all late-layer interventions produce no significant behavioral change ($p > 0.3$). This dissociation has a cross-lingual counterpart: the Llama ZH→ZH probe peaks at L13, the same layer where steering succeeds. Mid-layers thus serve a dual role as the site of cross-lingual alignment (Muller et al., 2021; Wendler et al., 2024) and the locus where conflict signals acquire causal force. The EN→EN peak at L28 is best understood as a diagnostic readout, a linearly readable trace of a decision that has already been made upstream. A practical implication follows: the probe peak layer is not the causal layer, and intervention experiments that target only the highest-accuracy layer risk missing the computationally relevant site. Cross-layer sweeps, as conducted here, are necessary to locate the true decision point. Steering failed in both Qwen (L14, 39%; $n = 131$, best $p = 0.785$) and Gemma (L17, 40%; $n = 171$, best $p = 0.448$), with zero directional flips in both cases. The Gemma result is particularly informative: its 88.9% probe accuracy—the highest among all three models—rules out probe quality as a confound, since the direction that best separates conflict labels carries no causal influence despite being highly discriminative. This establishes that the causal influence of the probe direction is architecture-dependent, not a general property of mid-layer computation.

Conflict signals are encoded in a general cross-lingual space. The ZH→EN transfer advantage is consistent across all three architectures (+7.3, +7.2, and +6.6 pp for Llama, Qwen, and Gemma respectively), and the XNLI baseline comparison (+6.0, +6.7, and +7.8 pp) is consistent with the conflict task asymmetry in magnitude across all three models. The asymmetry is therefore not conflict-specific; conflict detection piggybacks on the model's pre-existing cross-lingual alignment structure. This interpretation connects to the linguistic affinity effects reported by Zhao et al. (2026): the CLEAR benchmark found that conflict resolution preferences vary by language pair, a pattern that may inherit from the quality of underlying representational alignment documented by Kargaran et al. (2024) and Wendler et al. (2024). Any cross-lingual probing study must account for this baseline asymmetry before attributing transfer differences to task-specific phenomena.

Direct evidence for a shared cross-lingual subspace. The matching asymmetry magnitudes between conflict and XNLI tasks provide indirect evidence for a shared subspace; we test this directly via cross-task language probe transfer. A language identity probe (EN vs. ZH) trained on conflict task activations and evaluated on XNLI activations achieves ≥ 0.96 accuracy at most layers across all three models, with the reverse direction (XNLI→conflict) reaching ≥ 0.98, confirming that both tasks encode language identity in a common representational subspace. At Llama L13—the layer where steering succeeds—conflict→XNLI transfer is 0.998 [0.993, 1.000]. In Gemma, however, mid-layer transfer accuracy is reduced but

not absent at layers 20–22, reaching a minimum of 0.730 [0.653, 0.828] at L21—still well above chance (0.5) but notably lower than surrounding layers. This dip at L21 does not extend to L17, where cross-task transfer remains near-perfect (0.994) despite steering failure—suggesting that cross-lingual subspace disruption is not the sole factor behind negative steering outcomes.

Typological distance and asymmetry magnitude. The near-zero asymmetry for EN-DE ($\Delta = +1.2$ pp, CI $[-4.8, +8.1]$, n.s.) contrasts with the consistent +6–8 pp for EN-ZH across all three models, consistent with the hypothesis that asymmetry magnitude reflects the degree of representational alignment between the source and target languages. With only two language pairs, we report this as an observation rather than a tested claim. Notably, EN-DE token overlap (per-sample Jaccard 14.8%) is approximately $4\times$ higher than EN-ZH (3.8%), yet the EN↔DE transfer is near-symmetric, further ruling out shared tokens as a driver of the observed EN-ZH asymmetry.

Relation to interpretability without actionability. Basu et al. (2026) reported that steering failed in clinical NLP tasks despite near-perfect probe accuracy (98.2% AUROC). Our results present a partial contrast: mid-layer steering in knowledge conflict tasks succeeds with a weaker probe signal. A key difference is layer selection. Basu et al. evaluated steering at layers selected for probe accuracy; our analysis shows that the layer with highest probe accuracy (L28) also fails to steer, and causal influence is found only at L13. Task characteristics may also contribute: knowledge conflict resolution involves a discrete choice between two competing answer sources, which may concentrate causal structure at a single computational stage. We frame this as a difference in causal influence rather than full causal control, since steering shifts resolution probabilities without fully determining the outcome.

A suggestive MLP trend. MLP sublayer probes for EN→ZH show positive conflict-task gains (Qwen +3.84 pp, Llama +5.21 pp) while the XNLI baseline yields negative or zero gains, but both confidence intervals include zero, and this pattern should be interpreted cautiously pending larger-scale verification.

Scope and forward look. Our analysis covers one language pair (EN-ZH), three architectures for steering, and confirmed causal effects in Llama only (see §6). Future work should explore diverse language pairs and alternative steering methods that may capture non-linear causal structure missed by single-direction perturbation.

6 Conclusion

This work investigated whether knowledge conflict signals in multilingual LLMs are detectable across languages and whether detected signals causally influence conflict resolution. Through residual stream probing on Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-2-9B-IT, we showed that conflict signals are linearly detectable both within and across languages, with balanced accuracies of 77.6%, 75.0%, and 76.2% on English within-language detection and all cross-lingual transfer directions significantly exceeding chance (binomial $p < 0.001$). The observed transfer asymmetry (ZH→EN exceeding EN→ZH by approximately 6–7 pp) is consistent across all three architectures and is consistent with the XNLI non-conflict baseline in magnitude and direction for all models, establishing that this pattern reflects the general cross-lingual representational structure of English-dominant models rather than conflict-specific encoding.

A central finding is the dissociation between detection and decision. Probe accuracy peaks at late layers (L28, 88% depth), but representation steering succeeds only at a mid-layer (L13, 41% depth; Fisher $p = 0.002$), and all late-layer interventions fail. This result demonstrates that linear detectability does not imply causal influence, and that conflict resolution occurs upstream of where the outcome becomes linearly readable. For probing-based intervention research, this dissociation implies that cross-layer sweeps are necessary to identify the computationally relevant site, as targeting the probe peak alone can miss the true decision point.

Several directions remain open. Steering failed in both Qwen (L14; $n = 131$, best $p = 0.785$) and Gemma (L17; $n = 171$, 88.9% probe accuracy, best $p = 0.448$), with zero directional flips despite Gemma's probe accuracy exceeding Llama's, confirming that the causal influence of the probe direction is architecture-specific. Future work should explore alternative intervention methods, typologically diverse language pairs, and feature-level decomposition via sparse autoencoders.

Limitations

Our study has three main limitations. First, we evaluate only one language pair (English-Chinese). While the consistent patterns across three architecturally distinct models are encouraging, typologically diverse language pairs (e.g., English-German, English-Japanese) are needed to assess the generality of our findings. Second, causal steering succeeded only in Llama at L13. Gemma L17 steering failed despite the highest probe accuracy among all three models (88.9%, $n = 171$), ruling out probe quality as a confound and constituting an informative negative that confirms the architecture-specificity of the effect. Qwen L14 ($n = 131$, probe accuracy 71.0%) was also negative with zero directional flips. Third, the effective dataset size (966 to 1,590 samples per model after parametric knowledge verification) is constrained by the requirement that each model must correctly answer the question without context. This verification step is necessary to ensure genuine knowledge conflicts, but it limits the statistical power of our analyses, particularly for stratified comparisons such as the COMET tercile ablation. Fourth, all probing results are based on a single train/test split; while bootstrap confidence intervals capture sampling variability within this split, future work should verify stability across multiple random seeds. Fifth, Qwen3-8B serves as both the translation model and an experimental subject; while the COMET-based quality control and the Llama/Gemma cross-validation mitigate this concern, a fully independent translation model would strengthen the Qwen-specific results.

A Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.

B Fine-Grained Causal Boundary Analysis

Table 4 reports steering results across the full layer gradient from an independent replication ($N = 201$, Llama-3.1-8B-Instruct, RTX PRO 6000).

Table 4: Causal boundary steering results (independently sampled, $N = 201$, Llama-3.1-8B-Instruct). Bonf. Sig. αs: number of α values reaching significance after Bonferroni correction out of 6 tested.

Layer	Depth	Probe Acc	Best α	Best Fisher p	Bonf. Sig. αs	Result
L12	38%	91.75%	+20	0.085	0/6	—
L13	**41%**	**90.00%**	**+10**	**< 0.001**	**2/6**	✓✓
L14	44%	91.75%	+10	0.002	1/6	✓
L16	50%	91.50%	+10	0.005	1/6	✓
L18	56%	90.75%	+5	0.010	0/6	—
L28	88%	85.25%	+10	0.284	0/6	—

C Matched-Condition Steering Results

Table 5 reports steering results under matched conditions: all three architectures are tested at comparable relative depths (∼36–43%) using the same probe training methodology and six α values in $[-20, +20]$. Despite the highest probe accuracy among the three models (88–89%), Gemma produces zero directional flips, ruling out probe quality as a confound for the Llama-specific positive result. The full Llama causal gradient ($N = 201$) is in Appendix B.

Table 5: Matched-condition steering results. Probe: EN→EN balanced accuracy at the target layer. Best p: minimum Fisher exact p-value across six tested α values. Bonf.: α values surviving Bonferroni correction out of 6. Flips: total directional transitions (p→c + c→p).†

Layer	Depth	Probe (%)	Best p	Bonf.	Flips
Llama-3.1-8B-Instruct ($N = 136$; 32 layers)					
L13	**41%**	**76.1**	**< 0.001**	**2/6**	**9**
Qwen3-8B ($N = 131$; 36 layers)					
L13	36%	71.8	0.168	0/6	0
L14	39%	71.0	0.785	0/6	0
L15	42%	72.9	0.617	0/6	0
Gemma-2-9B-IT ($N = 171$; 42 layers)†					
L16	38%	88.0	0.067	0/6	0
L17	40%	88.9	0.448	0/6	0
L18	43%	89.5	0.219	0/6	0

† Gemma v2: re-run after correcting a label assignment bug in v1 data preparation.

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